# Multi-Granularity Aggregation Network for Remote Sensing Few-Shot Segmentation

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***Abstract*— Few-shot semantic segmentation (FSS) aims to segment a query image using a limited number of densely annotated support images from the same category. Most existing conventional FSS methods are tailored for coping with images from natural scenarios. Unlike natural images, remote sensing images usually have a similar background context among the support and query image pairs, and more severe intraclass inconsistency exists due to overhead shooting views. However, facing such realistic and challenging remote sensing FSS tasks, the existing methods seldom consider these intrinsic characteristics from a unified viewpoint, thus leading to inferior results. To solve the above dilemma, we propose a multi-granularity aggregation network (MGANet) to progressively capture multi-granularity discriminative information, for tackling the remote sensing FSS task. Specifically, MGANet consists of a multi-granularity similarity (MGS) module and an adaptive multiprototype aggregation (AMPA) module. To fully utilize background context, MGS extracts multi-granularity support and query feature maps from the backbone network to calculate a holistic correlation by incorporating the background information. Next, to alleviate the intraclass inconsistency of remote sensing images, AMPA decomposes the support foreground region into mainstay and auxiliary subregions by the guidance of reverse prediction on support features, thus generating three types of prototypes by masked average pooling (MAP) on these paired features and masks. Furthermore, these multiprototypes are collaboratively interacted with the query features to pursue reinforced discriminative features, relying on prototype-aware slot attention (PASA). Extensive experiments on iSAID-$5^i$ and LoveDA-$2^i$ demonstrate well the superiority of the proposed MGANet. The source code is available at https://github.com/CVL-hub/MGANet/.**

***Index Terms*— Background context, few-shot segmentation (FSS), intraclass inconsistency, multi-granularity, remote sensing, slot attention.**

## I. Introduction

SEMANTIC segmentation is crucial for assisting in interpreting remote sensing images. It has been extensively studied for many practical applications, such as land-cover mapping [1], [2], land use [3], forest management [4], [5], urban planning [6], [7], and environmental protection [8], [9]. Because of advancements in deep learning, semantic segmentation for remote sensing images has made significant progress in recent decades [10], [11], [12], [13], [14]. However, training these deep learning models requires a large amount of pixel-level annotations, which is time-consuming and labor-intensive. Although weakly supervised methods can alleviate the cost associated with data annotation to some extent, the performance of these models tends to decrease correspondingly. Besides, these models have poor generalization ability on novel tasks—tasks that need to segment objects from novel categories, while there is no data available during training for these categories—thus leading to inferior results and even failure of the model. Typically, the conventional models need to integrate new data and retrain the model to handle new tasks.

Humans can quickly grasp a novel concept from only a few examples; e.g., by solely seeing one image of a zebra, a child can recognize a different zebra in the wild under various appearances. As such, researchers propose the task setting of few-shot learning (FSL) [15] to mimic the aforementioned human capability. The goal of FSL is to enable models to learn how to learn, allowing them to discern differences and similarities from a small amount of data. Consequently, when encountering unseen categories during testing, the model can still effectively distinguish them. Few-shot segmentation (FSS) is the special application of FSL in the domain of semantic segmentation [16]. When provided with a query image to be segmented, FSS models aim to classify each pixel into predefined semantic categories, by relying on support images and their annotations.

Traditional FSS methods can be broadly divided into two categories, i.e., prototype-based methods and correlation-based methods. The former compresses the support feature map into a single and/or several vectors to obtain its class representation [17], [18], [19], [20], [21], [22], [23], [24], [25], which is then matched with the query feature to generate a prediction. The latter establishes pixel-to-pixel correlations through attention mechanisms [26], [27], [28] and/or parameter-free metric functions [29], [30], [31] (e.g., cosine similarity). However, these methods yield inferior results when faced with complex remote sensing scenarios. As in Fig. 1(a), they misclassify "ground" and "car" in the background as foreground "airplane" and produce incomplete segmentation of the wings

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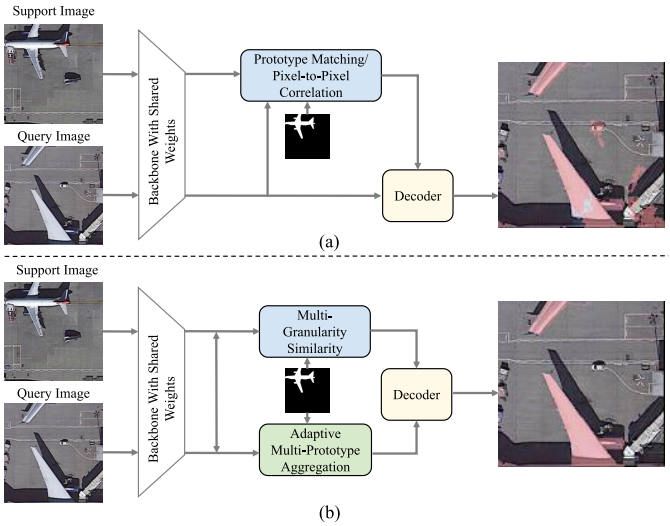



Fig. 1. Comparisons of the existing methods and our MGANet. (a) Existing methods, which solely rely on prototype matching and/or pixel-to-pixel correlation calculation, usually yield inferior segmentation results. They misclassify the "ground" and "car" in the background as foreground "airplane" and produce incomplete segmentation with respect to the foreground airplane "wings." (b) By contrast, our MGANet well addresses background object misclassification and incomplete foreground segmentation issues. This is because our MGANet considers the intrinsic characteristics of remote sensing images from a unified viewpoint. MGANet integrates an MGS module and an AMPA module to alleviate both background interference and intraclass inconsistency, respectively. Zoom in for details.

in the foreground. This is because these models seldom consider the intrinsic characteristics—i.e., the following hold: 1) a similar background context usually appears among the support and query image pairs and 2) more severe intraclass inconsistency exists due to overhead shooting views—of remote sensing images from a unified viewpoint. At first, these methods usually do not consider the relationship between the support background and the query background. Since the remote sensing images of the same category usually have similar background context—e.g., airplanes usually park at airports; as such, airplane class has similar background context of airports—these background contexts can, thus, be leveraged to obtain additional segmentation cues. In addition, due to overhead shooting views of remote sensing images, severe intraclass inconsistency is observed in remote sensing images. As in Fig. 1, the query image only contains the wings of an airplane, while the support image has a whole airplane object. This leads to a distribution difference between the whole airplane and its wings. The aforementioned methods typically rely on a unidirectional similarity calculation from support to query [16], [17], [18], [24], [32] and directly align support foreground features with the query features, ignoring the inconsistency between support foreground and the query foreground [22], [23], [33], [34]. When the segmentation target is the wings, adopting the corresponding wing subregion from the support image for guidance is more effective than using the entire airplane. In other words, support foreground subregions that are consistent with the query foreground can enhance segmentation accuracy.

To solve the above challenges, we propose a multi-granularity aggregation network (MGANet) to progressively leverage the support background information and the detailed support subregion information, for tackling the remote sensing FSS task. As in Fig. 1(b), our unified framework can well tackle complex remote sensing scenarios. Specifically, we first propose a multi-granularity similarity (MGS) module, which extracts support and query features under the same resolution from multiple backbone layers. Then, different from the existing methods that remove the support background, we preserve them without filtering out the support background and directly calculate the similarity matrices between the query and support features. In this way, the background context information is well utilized for similarity calculation. The similarity maps are then obtained by taking matrix multiplication of the support mask and the similarity matrices. Finally, we concatenate these multi-granularity features along the channel dimension and pass them through a single convolutional layer, for achieving the holistic multi-granularity features. By incorporating the support background, MGS can achieve accurate segmentation by filtering out query features similar to the support background and retaining desirable features with respect to support foreground.

Furthermore, to adaptively utilize support subregion information, we propose an adaptive multiprototype aggregation (AMPA) module. AMPA decomposes the support foreground into mainstay and auxiliary subregions, to achieve a desirable match while reducing the impact of noise. AMPA consists of a prototype generation (PG) module and a prototype-aware slot attention (PASA) module for feature enhancement. PG module generates an original support prototype and adopts it to predict an initial rough query object, thus producing a pseudoquery prototype through a masked average pooling (MAP) operation. Then, we use this query prototype to reversely predict the support object, which results in a pseudosupport mask. The common foreground region between this pseudosupport mask and the ground-truth support mask reflects the shared semantic (mainstay) information between the support and the query, while the remaining region serves as auxiliary information. The above task-aware mainstay and auxiliary masks are further used to generate the mainstay and auxiliary prototypes by MAP, respectively. In addition, PASA is an object-centric attention mechanism [35]. Here, we take the query features and the group of the original prototype, mainstay prototype, and auxiliary prototype as input to PASA for pursuing reinforced discriminative query features. Through iterative interaction, PASA gradually activates features similar to a given prototype to enhance the features.

In summary, our main contributions are outlined as follows.

1) We propose a unified MGANet for addressing the remote sensing FSS task. It can well tackle complex remote sensing scenarios by capturing the holistic correlation between multi-granularity support and query features and utilizing detailed support foreground subregions to enhance the discriminative information in the query image.
2) We propose an MGS module that locates the query object area by computing similarity maps at multiple granularities. It benefits from leveraging support

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background information to engage in similarity calculation, which results in informative similarity score vectors, and, thus, achieving accurate segmentation in remote sensing scenarios.

3) We propose an AMPA module, which first generates multiple prototypes that aggregate information from different regions of the support foreground. Then, PASA is further utilized to enhance the query features. To the best of our knowledge, this is the first try to do this in the remote sensing FSS domain.
4) We conduct extensive experiments on two golden remote sensing FSS datasets, iSAID-$5^i$ and LoveDA-$2^i$, to validate the effectiveness of MGANet. The results demonstrate that the MGANet achieves the best performances on both datasets.

The remainder of this article is organized as follows. Section II reviews significant works on semantic segmentation, FSL, and FSS. Section III provides a detailed illustration of the proposed MGANet. Section IV presents the experimental results and ablation studies. Finally, Section V presents the conclusion.

## II. Related Work

### A. Semantic Segmentation

Semantic segmentation, which assigns a unique category to each pixel in an image, is a critical task in computer vision. Long et al. [36] propose pioneer work in this field by replacing the final fully connected layer of CNNs with fully convolutional layers, enabling the network to accommodate inputs of any size. U-Net [37] introduces an encoder–decoder architecture with skip connections to integrate features from different levels. PSPNet [38] emphasizes the importance of contextual information, designing the pyramid pooling module to merge features at various scales. The DeepLab [39] models achieve exceptional results in semantic segmentation using techniques, such as atrous convolution and atrous spatial pyramid pooling.

Deep-learning-based technologies also receive significant attention from researchers in the field of semantic segmentation for remote sensing imagery [40], [41], [42], [43], [44]. Li et al. [45] advocate using boundary information to enhance multiscale features for semantic segmentation of high-resolution remote sensing images. Kampffmeyer et al. [46] perform uncertainty modeling on CNNs and conduct both qualitative and quantitative analyses in remote sensing scenarios, resulting in high segmentation accuracy for small-sized objects. Multimodal data can represent target objects from multiple perspectives. Ma et al. [47] fuse multilevel multimodal data, thereby achieving better semantic segmentation results. Zhang et al. [48] are the first to explore federated learning in remote sensing scenarios. To address the issue of heterogeneous data among clients, they design a prototype-based matching method to constrain the learning of local prototypes and the aggregation of global prototypes. Recently, some studies have also applied advanced models or architectures to remote sensing image segmentation. For instance, RSAM-Seg [49] and SSRS [50] leverage the powerful generalization capability of SAM [51] to enhance segmentation accuracy. Ma et al. [52] combine the ability of Vision Mamba [53] to model long-range dependencies with linear time complexity and design the $RS^3$Mamba framework for remote sensing data. However, these fully supervised methods and models struggle to generalize to unseen domains and categories.

### B. Few-Shot Learning

FSL is designed to help models acquire highly generalizable knowledge from a few samples. This cost-effectiveness has drawn significant research interest in recent years. Current FSL methods can be broadly categorized into three types. Some methods follow the transfer-learning paradigm [54], [55], i.e., training on base classes and then fine-tuning on novel classes. The second type is the metric-based approach [56], [57], [58], which learns a general metric to compare support and query samples, ultimately determining whether they belong to the same category. The last is the meta-learning approach [59], [60], [61], [62], which follows the meta-learning paradigm. During training, they construct a series of episodes (subtasks) to learn the optimization process for updating model parameters. In addition, some methods utilize language guidance. For instance, Ji et al. [63] use GPT [64] to generate rich prompts for CLIP [65] and then embed the outputs of CLIP's text encoder into visual prototypes. Cheng and Yang [66] align visual features with textual features to correct classification biases within a single modality.

While prior research on FSL predominantly focuses on classification tasks, our study addresses the more challenging task of few-shot semantic segmentation (FSS) in the context of remote sensing images.

### C. Few-Shot Semantic Segmentation

FSS inherits the main settings of FSL. As a dense prediction task, FSS aims to assign semantic labels to every pixel in an image, even with only a few annotated examples available per class. Current FSS methods can be broadly divided into two categories, i.e., prototype-based and correlation-based. The former compresses the support foreground information into one or more prototypes and then compares them with the query features to produce the prediction results. For instance, PMMS [19] uses the EM algorithm to generate various prototypes with different distributions to alleviate the semantic ambiguity problem. ASGNet [17] employs a superpixel clustering algorithm to aggregate similar pixel vectors in the support features and dynamically replaces the vector most similar to each query pixel in the corresponding position for adaptive matching. SSP [22] proposes a self-support matching approach, matching the query prototype with the query features, which differs from previous methods that use query prototypes [23], [33]. However, this strategy of fusing information from multiple support pixels inevitably leads to the loss of local information, thus leading to only rough segmentation of the query target region.

In addition, correlation-based methods achieve better results. They predict the query mask by calculating the pixelwise correlation between paired support and query images.

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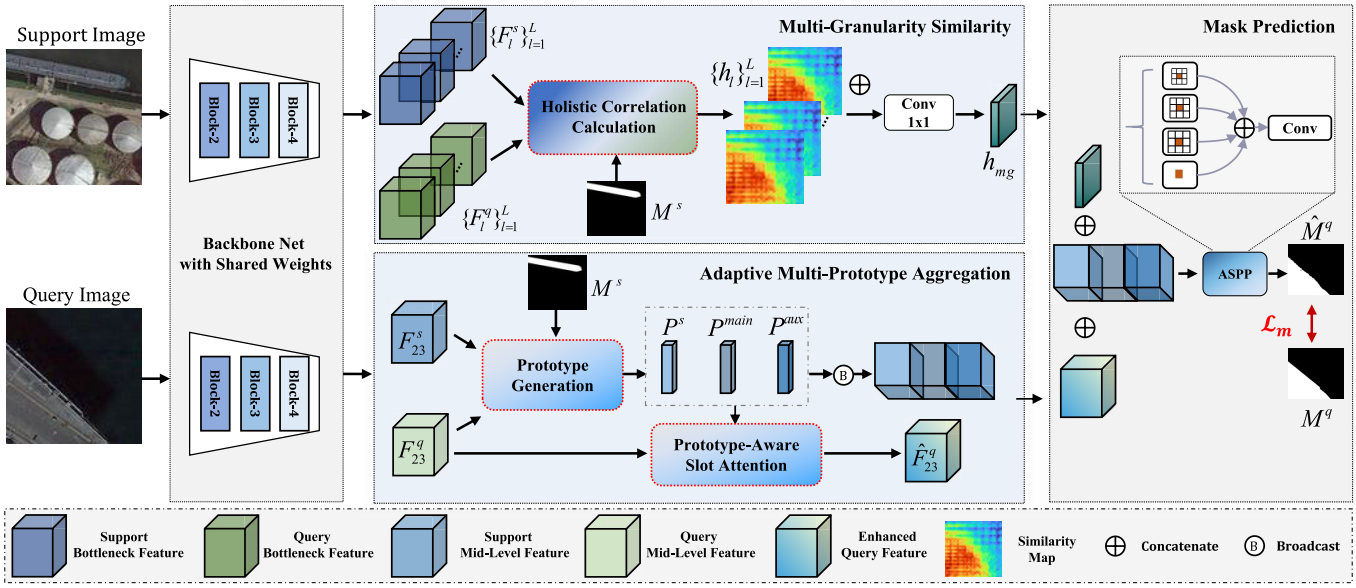



Fig. 2. Architecture of MGANet. We illustrate the training phase under the one-shot setting. Our network mainly consists of two modules: the MGS module and the AMPA module. MGANet takes support and query image pair as input. Then, bottleneck features ($\{(F_l^s, F_l^q)\}_{l=1}^L$ ) and mid-level features ($F_{23}^s$ and $F_{23}^q$) are extracted for them. These bottleneck and mid-level features are further fed into MGS and AMPA modules, thus leading to more desirable features. Finally, all these features are leveraged for the query image prediction by relying on an ASPP-based decoding module. Best viewed in color.

Wang et al. [67] and Zhang et al. [68] use graph attention mechanisms to establish the correlation between support foreground features and query features. MSI [69] addresses the challenge of segmenting difficult samples in FSS by creating two complementary correlation maps using all support features and support foreground features with query features. Based on the transformer architecture, HDMNet [26] establishes a cascade matching between support and query and applies KL divergence supervision between adjacent correlation maps to achieve fine-grained segmentation. To address the local noise introduced by pixelwise matching, FECANet [70] aggregates neighborhood information on the feature map to increase the receptive field of each pixel, thereby capturing more contextual information. Furthermore, several methods utilize textual information to assist in FSS tasks [71], [72], [73]. Specifically, Wang et al. [74] employ CLIP [65] to generate unbiased prior information to locate the target class.

By contrast, our MGANet considers the similar background context among the support and query image pairs and more severe intraclass inconsistency due to overhead shooting views from a unified viewpoint. We introduce the MGS module and the AMPA module to progressively leverage the fine-grained semantic correlation information and the detailed support subregion information, for tackling the remote sensing FSS task.

## III. Multi-Granularity Aggregation Network

### A. Task Description

Remote sensing FSS aims to learn and transfer knowledge from a limited number of densely annotated support samples to segment similar objects in query images. To achieve this, we create a training dataset $D_{\text{train}}$ and a test dataset $D_{\text{test}}$ with two nonoverlapping sets of categories, i.e., $C_{\text{base}}$ and $C_{\text{novel}}$, thus ensuring that $C_{\text{base}} \cap C_{\text{novel}} = \emptyset$. The model's generalization ability is assessed on $D_{\text{test}}$ after being trained on $D_{\text{train}}$, without a fine-tuning stage. Similar to most FSS methods, we utilize the meta-learning paradigm, where a series of episodes (subtasks) are sampled from $D_{\text{train}}$ to mimic the few-shot conditions during meta-testing. Each episode comprises a query set $Q = \{(I^q, M^q)\}$ and a support set $S = \{(I_i^s, M_i^s)\}_{i=1}^k$ that share the same semantic category $c$. Here, $I^*$ and $M^*$ represent the original image and its binary mask, respectively. During the meta-learning phase, the support set $S$ and query image $I^q$ are input into the FSS model for predicting the mask $\hat{M}^q$. While training, the ground-truth mask $M^q$ is used to compute the loss, but it is not available during the meta-testing phase. Consistent with previous work, each episode includes only one semantic category. In addition, we focus on the model's performance under $k = 1$ and $k = 5$, respectively.

### B. Method Overview

Fig. 2 illustrates the architecture of our MGANet, which consists of MGS, AMPA, and mask prediction module. Given an episode $(S, Q)$, at first, we extract bottleneck features $\{(F_l^s, F_l^q)\}_{l=1}^L$ of the same resolution for support and query images from blocks 2 to 4 of a backbone network pretrained on ImageNet [75]. These features are then fed into the MGS module, which calculates their multi-granularity holistic correlations, and the holistic features are further processed through a $1 \times 1$ convolution.

On the other hand, our AMPA module first extracts the mid-level features $F_{23}^s$ and $F_{23}^q$ of support and query image, respectively. Together with the support ground-truth mask, these features are fed into the prototype generator to generate multiple support prototypes based on a batch of episodes.

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Each batch of episodes generates its prototypes, which are recalculated under a new batch. We then employ PASA, which uses an object-centric approach to activate query features that are highly correlated with these prototypes, while suppressing those that are not. Finally, the MGS map $h_{\text{mg}}$, the prototype set $P = \{P^s, P^{\text{main}}, P^{\text{aux}}\}$, and the enhanced query features are input into the ASPP decoder to generate the final prediction.

### C. Multi-Granularity Similarity

MGS mainly contains the holistic correlation calculation (HCC) and afterward convolutional compression operation. For HCC, the support/query feature set $\{(F_l^s, F_l^q)\}_{l=1}^{L}$ is obtained by performing interpolation and convolution operations on the multiscale and multilevel bottleneck features, thus making them have the same resolution. Next, the objective is to locate the target region within the query features. As in Fig. 3, this process is achieved by computing the cosine similarity between the support and query features pixel-by-pixel. Specifically, taking a specific pair of features, $F_l^s \in \mathbb{R}^{C_l \times H \times W}$ and $F_l^q \in \mathbb{R}^{C_l \times H \times W}$, from the feature set as an example, the element of their similarity matrix $S_l \in \mathbb{R}^{HW \times HW}$ is derived as follows:

$$S_l(i, j) = \frac{\hat{F}_{\text{li}}^q \hat{F}_{\text{lj}}^{s\top}}{\|\hat{F}_{\text{li}}^q\| \|\hat{F}_{\text{lj}}^s\| + \epsilon} \tag{1}$$

where $\hat{F}_l^s \in \mathbb{R}^{HW \times C_l}$ ($\hat{F}_l^q \in \mathbb{R}^{HW \times C_l}$) is reshaped by $F_l^s \in \mathbb{R}^{C_l \times H \times W}$ ($F_l^q \in \mathbb{R}^{C_l \times H \times W}$), and $\hat{F}_{\text{li}}^s$ and $\hat{F}_{\text{lj}}^q$ are the $i$ row and $j$ row of $\hat{F}_l^q$ and $\hat{F}_l^s$, respectively. $\epsilon$ is set to $1\text{e}-5$ for balancing the model training. Furthermore, softmax normalization is performed along each row of $S_l$, which helps to weight the relative importance of the support pixels with respect to each query pixel. We rewrite the normalized matrix still as $S_l$, and this normalization calculation is as follows:

$$S_l = \text{softmax}(S_l) \in \mathbb{R}^{HW \times HW}. \tag{2}$$

Then, to filter out similarity scores corresponding to the background regions in the support image, we perform a matrix multiplication operation between the support mask $M_l^s \in \mathbb{R}^{H \times W}$ and the similarity matrix $S_l$. As such, we obtain similarity vector $h_l \in \mathbb{R}^{HW \times 1}$ by this multiplication as follows:

$$h_l = S_l \hat{M}_l^s \in \mathbb{R}^{HW \times 1} \tag{3}$$

where $\hat{M}_l^s \in \mathbb{R}^{HW \times 1}$ is reshaped by $M_l^s \in \mathbb{R}^{H \times W}$. Previous works [16], [29], [30], [31] remove the support background information before computing the similarity matrix ($S_l$)—termed preremoving methods. They focus on exploiting foreground relationships, ignoring that additional segmentation cues can be obtained from the background. By contrast, our MGS leverages support background information to engage in similarity calculation, which results in informative similarity score vectors (each row of $S_l$) by the normalization operation (2). Finally, we also remove the support background information based on the informative $S_l$. Our scheme is termed a postremoving operation. Our simple postremoving operation for capturing background context information can bring significant performance gains (see Table IX).

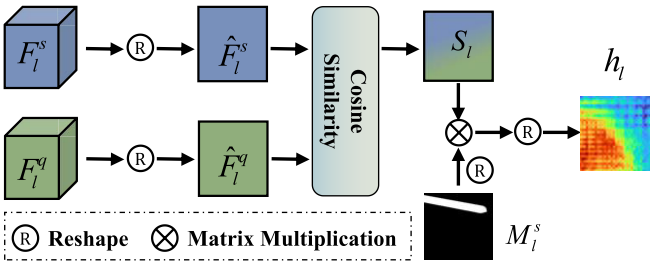



Fig. 3. Details of HCC. The probability of each query pixel belonging to the foreground is obtained by a weighted sum of all the support pixels and support mask.

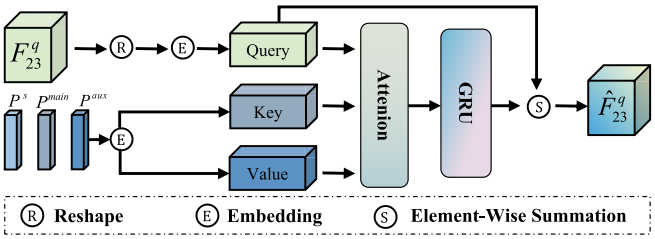



Fig. 4. Details of PASA. We denote the linearly transformed query features as $Q$, and the linearly transformed prototype features as $K$ and $V$.

Next, we perform min–max normalization on $h_l$ to scale its values between 0 and 1

$$h_l = \frac{h_l - \min(h_l)}{\max(h_l) - \min(h_l) + \epsilon}. \tag{4}$$

Finally, we reshape $h_l$ to the shape of $1 \times H \times W$, also denoted as $h_l$. In this way, we obtain the similarity map $h_l$ for the query feature from the $l$th layer. We perform the same operations on the remaining support/query feature pairs in the feature set. This results in MGS maps $\{h_l\}_{l=1}^{L}$, which are concatenated along the channel dimension and then passed through a $1 \times 1$ convolutional operation, thus leading to the final fused map as follows:

$$h_{\text{mg}} = \text{Conv}_{1\times1}(h_1 \oplus h_2 \oplus \cdots \oplus h_L) \tag{5}$$

where the number of output channels is set to 64, i.e., $\text{Conv}_{1\times1} : \mathbb{R}^{L \times H \times W} \Rightarrow \mathbb{R}^{64 \times H \times W}$, and $\oplus$ is the concatenating operation.

### D. Adaptive Multiprototype Aggregation

In the following, we introduce the PG module and the PASA module in detail.

*1) Prototype Generation:* Inspired by [16] and [76], we extract the intermediate support and query features from the backbone network, concatenate them along the channel dimension, and use a $1 \times 1$ convolution to compress the channel dimension to $C$

$$\begin{aligned} F_{23}^s &= \text{Conv}_{1\times1}(\bar{F}_2^s \oplus \bar{F}_3^s) \\ F_{23}^q &= \text{Conv}_{1\times1}(\bar{F}_2^q \oplus \bar{F}_3^q). \end{aligned} \tag{6}$$

In (6), $\bar{F}_2^*$ and $\bar{F}_3^*$ represent the features of block2 and block3, respectively, while $F_{23}^*$ denotes the fused feature that combines information from $\bar{F}_2^*$ and $\bar{F}_3^*$. Fig. 5 illustrates the flowchart

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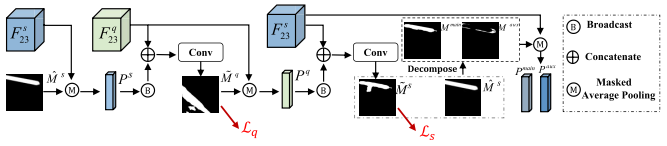



Fig. 5. Flowchart of the PG process.

of PG. Next, we can directly use the MAP operation to obtain the support prototype

$$P^s = \text{MAP}(F_{23}^s, \hat{M}^s) \tag{7}$$

where $\hat{M}^s$ denotes the downsampled support mask, and its resolution matches that of $F_{23}^s$. $P^s$ compresses all the support foreground pixel information.

Next, we can use $P^s$ to match with the query feature, generating a pseudoquery mask $\tilde{M}^q$. In detail, we use broadcast $P^s$, so that its shape changes from $C \times 1$ to $C \times H \times W$. Then, we concatenate $P^s$ with $F_{23}^q$ and obtain a pseudoquery mask through several convolutional layers. The above steps can be expressed as follows:

$$\tilde{M}^q = \text{Convs}(F_{23}^q \oplus \text{Broadcast}(P^s))). \tag{8}$$

After that, the query prototype can be obtained from the pseudoquery mask $\tilde{M}^q$ and the query feature $F_{23}^q$

$$P^q = \text{MAP}(F_{23}^q, \tilde{M}^q). \tag{9}$$

Under the guidance of the query prototype, we can reversely predict the pseudosupport mask $\tilde{M}^s$

$$\tilde{M}^s = \text{Convs}(F_{23}^s \oplus \text{Broadcast}(P^q)). \tag{10}$$

The regions in the support image that have a distribution consistency with the query object significantly contribute to the segmentation, while other regions may contribute little. By decomposing these two types of information from the support image, we can achieve a more precise matching. Specifically, $\tilde{M}^s$ is predicted using the query prototype, which is derived from the support prototype's prediction on the query. Consequently, $\tilde{M}^s$ contains the common semantic information between the support and the query. Using the support ground truth $\hat{M}^s$ as a reference, we can decompose it as follows:

$$\begin{aligned} M_{(i,j)}^{\text{main}} &= \mathbf{1}[\hat{M}_{(i,j)}^s = \tilde{M}_{(i,j)}^s = 1] \\ M_{(i,j)}^{\text{aux}} &= \hat{M}_{(i,j)}^s - M_{(i,j)}^{\text{main}} \end{aligned} \tag{11}$$

where $(i, j)$ represents the spatial coordinates. $\mathbf{1}[x]$ equals 1 when $x$ is true, and 0 otherwise. Here, $M^{\text{main}}$ is the common region between $\tilde{M}^s$ and $\hat{M}^s$, reflecting the support subregions that are inconsistent with query. Thus, it aggregates the mainstay foreground information. $M^{\text{aux}}$ is the region within $\hat{M}^s$ but outside $\tilde{M}^s$, and thus, it aggregates the auxiliary foreground information and provides supplementary segmentation cues.

The corresponding prototypes based on mainstay and auxiliary masks (11) can also be obtained through MAP operation

$$\begin{aligned} P^{\text{main}} &= \text{MAP}(F_{23}^s, M^{\text{main}}) \\ P^{\text{aux}} &= \text{MAP}(F_{23}^s, M^{\text{aux}}). \end{aligned} \tag{12}$$

*2) Prototype-Aware Slot Attention:* To fully leverage the support prototype set $P = \{P^s, P^{\text{main}}, P^{\text{aux}}\}$ that aggregates information from different regions to enhance query features, we design a PASA module based on slot attention [35]. Slot attention is an attention-based mechanism focusing on different slots, and these randomly initialized slots can capture different concepts or representations from features after several iterations. As in Fig. 4, different from vanilla slot attention, we use the reshaped query feature's embedding $\hat{F}_{23}^q \in \mathbb{R}^{HW \times C}$ as the initialized slots (i.e., query $Q$) and the embeddings of the prototype set $P$ transformed by a linear transformation as the key $K$ and value $V$. The attention map $A$ can be obtained by performing matrix multiplication between $Q$ and $K$

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{C}}\right) \in \mathbb{R}^{HW \times 3}. \tag{13}$$

Then, the reinforced attention features $F_{\text{att}}$ is calculated as follows:

$$F_{\text{att}} = \text{AV} \in \mathbb{R}^{HW \times C}. \tag{14}$$

At each iteration, the attention features are updated using a GRU function. After $T$ iterations, the query features can aggregate information from the prototype set by weighting them based on their importance and relevance. The GRU iteration function is as follows:

$$\begin{aligned} \text{slots} &= \text{GRU}(\text{states} = \text{slots}, \text{inputs} = F_{\text{att}}) \\ \hat{F}_{23}^q &= \hat{F}_{23}^q + \text{slots} \end{aligned} \tag{15}$$

where $\hat{F}_{23}^q$ can be recursively utilized to generate $F_{\text{att}}$ based on (13) and (14) for the next GRU iteration. Finally, by reshaping $\hat{F}_{23}^q$ back to the original feature dimension, we obtain the enhanced features. Guided by the prototype set, prototype-aware slot attention can enhance features related to the specified prototype, thus leading to more precise segmentation results.

### E. Training Losses

The model decoder takes the MGS map $h_{\text{mg}}$, the prototype set $P$, and the enhanced query feature $\hat{F}_{23}^q$ as inputs to predict the query mask $\hat{M}^q$. Binary cross entropy (BCE) is used as the loss function to measure the difference between the prediction and the ground truth

$$\mathcal{L}_m = \text{BCE}(\hat{M}^q, M^q) \tag{16}$$

where $M^q$ represents the ground-truth binary mask of the query image. In addition, to generate more precise prototypes, we also impose BCE constraints on the query pseudomask $\tilde{M}^q$ and support pseudomask $\tilde{M}^s$

$$\begin{aligned} \mathcal{L}_q &= \text{BCE}(\tilde{M}^q, M^q) \\ \mathcal{L}_s &= \text{BCE}(\tilde{M}^s, M^s). \end{aligned} \tag{17}$$

Finally, the model's parameters are updated by simultaneously optimizing all the previously mentioned losses

$$\mathcal{L} = \mathcal{L}_m + \alpha\mathcal{L}_q + \beta\mathcal{L}_s. \tag{18}$$

Here, $\alpha$ and $\beta$ are hyperparameters for balancing corresponding losses.

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TABLE I

PRECISION RESULTS OF THE MODEL ON THE iSAID-$5^i$ BENCHMARK UNDER ONE-SHOT AND FIVE-SHOT SETTINGS ARE PRESENTED BELOW. THE EVALUATION METRICS INCLUDE mIoU FOR EACH FOLD, THE MEAN mIoU ACROSS ALL FOLDS (MEAN), AND FB-IoU. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

| Backbone | Method | 1-shot | | | | | 5-shot | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | fold0 | fold1 | fold2 | Mean(%) | FB-IoU(%) | fold0 | fold1 | fold2 | Mean(%) | FB-IoU(%) |
| VGG-16 | SADMNet [32] | 29.24 | 20.80 | 34.73 | 28.26 | - | 36.33 | 27.98 | 42.39 | 35.57 | - |
| | Jiang et al [82] | 33.40 | 35.13 | 43.28 | 37.27 | 57.72 | 37.33 | 36.08 | 44.52 | 39.31 | 59.01 |
| | SCL [18] | 46.88 | 31.74 | 44.64 | 41.09 | 60.88 | 47.75 | 31.98 | 43.45 | 41.06 | 60.69 |
| | PFENet [16] | 46.99 | 35.33 | 49.52 | 43.95 | 60.81 | **53.34** | 38.55 | 50.61 | 47.50 | 62.94 |
| | ASGNet [17] | 45.12 | 32.25 | 43.30 | 40.22 | 59.68 | 51.88 | 33.93 | 50.96 | 45.59 | 63.07 |
| | DCPNet [80] | 47.44 | 33.91 | 48.08 | 43.14 | 60.43 | 48.16 | 36.07 | 49.25 | 44.49 | 62.03 |
| | NTRENet [81] | 47.96 | 36.21 | **51.22** | 45.13 | 61.50 | 49.05 | 37.66 | 52.33 | 46.35 | 62.24 |
| | DMNet [21] | **50.64** | 38.59 | 46.32 | 45.18 | **63.10** | 52.09 | 40.54 | 50.39 | 47.67 | **63.70** |
| | Ours | 49.55 | **40.47** | 48.72 | **46.25** | 62.47 | 52.48 | **42.26** | **52.35** | **49.02** | 63.47 |
| ResNet-50 | SADMNet [32] | 34.29 | 22.25 | 35.62 | 30.72 | - | 39.88 | 30.59 | 45.70 | 38.72 | - |
| | Jiang et al [82] | 35.11 | 37.06 | 42.24 | 38.14 | 58.28 | 38.19 | 37.77 | 43.94 | 39.97 | 59.13 |
| | SCL [18] | 49.08 | 35.61 | 48.15 | 44.28 | 61.66 | 50.69 | 35.64 | 48.72 | 45.02 | 62.63 |
| | PFENet [16] | 51.34 | 38.79 | 52.26 | 47.46 | 63.34 | 54.71 | 41.51 | 54.45 | 50.22 | 64.99 |
| | ASGNet [17] | 48.59 | 36.82 | 46.65 | 44.02 | 62.36 | 53.01 | 37.44 | 52.18 | 47.54 | 64.63 |
| | CyCTR [79] | 51.15 | 38.40 | 53.79 | 47.78 | 62.86 | 51.91 | 39.01 | 54.83 | 48.58 | 63.81 |
| | DCPNet [80] | 48.43 | 37.59 | 52.09 | 46.04 | 62.76 | 50.34 | 40.75 | 51.33 | 47.47 | 64.10 |
| | NTRENet [81] | 49.52 | 38.66 | 51.87 | 46.68 | 62.84 | 50.60 | 40.99 | 55.07 | 48.89 | 63.74 |
| | DMNet [21] | 54.45 | 40.68 | 53.60 | 49.58 | 64.46 | 57.67 | 41.06 | 55.28 | 51.14 | 65.81 |
| | SCCAN [83] | 52.24 | 38.60 | 52.84 | 47.89 | 63.26 | 52.75 | 38.95 | 54.26 | 48.65 | 62.23 |
| | $R^2$Net [84] | **56.81** | 39.85 | 49.02 | 48.56 | 62.65 | **60.47** | 41.43 | 50.24 | 50.71 | 65.70 |
| | Ours | 55.81 | **43.52** | **56.23** | **51.85** | **64.49** | 57.81 | **45.32** | **57.87** | **53.67** | **65.99** |
| ResNet-101 | SCL [18] | 47.59 | 36.90 | 45.21 | 43.23 | 61.71 | 48.94 | 38.01 | 46.21 | 44.39 | 60.93 |
| | PFENet [16] | 50.69 | 38.37 | 52.85 | 47.30 | 62.46 | 54.40 | 41.55 | 50.55 | 48.83 | 64.57 |
| | ASGNet [17] | 47.55 | 38.47 | 49.28 | 45.10 | 62.03 | 53.54 | 38.24 | 53.20 | 48.33 | 65.35 |
| | CyCTR [79] | 50.89 | 38.89 | 52.22 | 47.73 | 62.35 | 52.15 | 40.28 | 55.32 | 49.25 | 64.45 |
| | DCPNet [80] | 47.63 | 38.80 | 49.34 | 45.26 | 62.56 | 50.68 | 40.02 | 54.52 | 48.41 | 62.96 |
| | NTRENet [81] | 50.33 | 38.73 | 51.23 | 46.76 | 63.25 | 53.24 | 41.87 | 51.53 | 48.88 | 64.16 |
| | DMNet [21] | 54.01 | 40.04 | 53.57 | 49.21 | 64.03 | 55.70 | 41.69 | 56.47 | 51.29 | 65.88 |
| | SCCAN [83] | 52.71 | 38.76 | 53.45 | 48.21 | 62.72 | 53.85 | 39.11 | 54.56 | 49.17 | 64.27 |
| | $R^2$Net [84] | **57.74** | 39.76 | 48.45 | 48.65 | 63.38 | **58.83** | 40.58 | 49.69 | 49.70 | 63.79 |
| | Ours | 55.03 | **42.90** | **54.69** | **50.87** | **64.35** | 56.43 | **44.85** | **57.67** | **52.98** | **66.42** |

## IV. EXPERIMENTS

In our experiments, we use two datasets: iSAID-$5^i$ [77] and LoveDA-$2^i$ [78]. FB-intersection-over-union (IoU) and mIoU are used as evaluation metrics for model performance. First, we follow the BAM [24] setup by removing images containing novel classes from the training set and compare our MGANet with mainstream FSS models (Tables I and II). Next, we conduct ablation studies to assess the impact of each module and hyperparameter on the results. In addition, we further validate the superiority of our method under the non-BAM setting by labeling novel classes present in the training images as background.

### A. Settings

*1) Datasets:* iSAID-$5^i$ is constructed by Yao et al. [32] from the iSAID dataset [77], and it comprises 655 451 object instances across 15 categories within 2806 high-resolution images. It is a widely used standard dataset in remote sensing FSS. Specifically, the 15 categories are divided into threefold. The model is trained on any two of these folds (ten object categories) and tested on the remaining fold (five object categories).

LoveDA is a dataset designed for urban-to-rural domain adaptation, comprising 5987 images and 166 768 annotated semantic objects from three cities. Building on this, Bi et al. [21] create the LoveDA-$2^i$ benchmark. This dataset is more challenging due to the distribution shift in the sample data and the variation in object scales. After removing the complex background and no data categories, the remaining six categories are evenly split into threefold for cross validation.

*2) Evaluation Metrics:* In this article, we utilize two metrics, mIoU and FB-IoU, to evaluate the model's performance. The mIoU is calculated by computing the IoU for each foreground category and then averaging the results. In contrast, the FB-IoU treats all categories in the test set as a single foreground category, calculating the IoU between the foreground and the background, and averaging these values. MIoU is more crucial than FB-IoU, because in real-world scenarios, models encounter data from multiple classes of objects. We need a comprehensive evaluation of the model's performance across all classes to mitigate biases.

### B. Implementation Details and Parameters

Our model is implemented using the PyTorch framework, and all experiments are conducted on two NVIDIA GeForce RTX 3090 GPUs. The backbone networks (VGG-16, ResNet-50, and ResNet-101) use pretrained weights from ImageNet [75], and their parameters are frozen during training.

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TABLE II

RESULTS OF THE MODEL ON THE LOVEDA-$2^i$ DATASET UNDER ONE-SHOT AND FIVE-SHOT SETTINGS ARE PRESENTED BELOW. THE EVALUATION METRICS INCLUDE MIOU FOR EACH FOLD, THE MEAN MIOU ACROSS ALL FOLDS (MEAN), AND FB-IOU. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

| Backbone | Method | 1-shot | | | | | 5-shot | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | fold0 | fold1 | fold2 | Mean(%) | FB-IoU(%) | fold0 | fold1 | fold2 | Mean(%) | FB-IoU(%) |
| VGG-16 | SCL [18] | 15.31 | 21.43 | 23.89 | 20.21 | 29.56 | 14.84 | 22.39 | 20.85 | 19.36 | 29.22 |
| | PFENet [16] | 16.14 | 24.35 | 31.57 | 24.02 | 34.72 | 15.08 | 26.82 | 30.18 | 24.03 | 38.40 |
| | ASGNet [17] | 15.93 | 21.60 | 29.08 | 22.20 | 39.79 | 17.33 | 26.24 | 35.75 | 26.44 | 38.41 |
| | DCPNet [80] | 15.22 | 22.58 | 31.83 | 23.21 | 34.97 | 15.94 | 26.38 | 31.37 | 24.56 | 36.24 |
| | NTRENet [81] | 15.07 | 23.17 | 28.68 | 22.31 | 36.18 | 15.30 | 25.12 | 30.95 | 23.79 | 35.06 |
| | DMNet [21] | 19.71 | 26.23 | 30.43 | 25.46 | **45.78** | 25.02 | **35.51** | 33.62 | 31.38 | **51.70** |
| | Ours | **20.00** | **27.44** | **32.63** | **26.69** | 43.11 | **25.09** | 35.03 | **36.64** | **32.25** | 50.44 |
| ResNet-50 | SCL [18] | 15.14 | 20.45 | 25.00 | 20.20 | 24.60 | 14.25 | 21.09 | 23.65 | 19.66 | 24.69 |
| | PFENet [16] | 17.13 | 22.20 | 26.49 | 21.94 | 33.48 | 15.83 | 25.73 | 24.74 | 22.10 | 34.77 |
| | ASGNet [17] | 15.91 | 20.21 | 22.33 | 19.48 | 36.39 | 18.38 | 26.29 | 36.34 | 27.00 | 39.59 |
| | CyCTR [79] | 13.17 | 23.43 | 21.99 | 19.53 | 38.47 | 13.81 | 27.40 | 26.15 | 22.45 | 42.71 |
| | DCPNet [80] | 16.67 | 23.10 | 24.44 | 21.40 | 36.36 | 13.43 | 25.59 | 28.06 | 22.36 | 33.89 |
| | NTRENet [81] | 16.05 | 22.69 | 21.87 | 20.20 | 32.67 | 15.79 | 24.94 | 23.42 | 21.38 | 31.79 |
| | DMNet [21] | 19.29 | 25.52 | 31.53 | 25.45 | 43.40 | 24.62 | 33.80 | 33.12 | 30.51 | **50.93** |
| | Ours | **20.55** | **27.51** | **33.30** | **27.12** | **44.91** | **26.53** | **35.99** | **36.69** | **33.07** | 50.68 |
| ResNet-101 | SCL [18] | 15.62 | 17.87 | 25.54 | 19.68 | 31.03 | 14.96 | 20.26 | 24.02 | 19.75 | 31.57 |
| | PFENet [16] | 15.83 | 25.73 | 24.74 | 22.10 | 35.65 | 15.62 | 24.37 | 26.64 | 22.21 | 36.08 |
| | ASGNet [17] | 14.65 | 19.90 | 25.43 | 19.99 | 30.89 | 18.00 | 25.43 | 36.00 | 26.48 | 39.26 |
| | CyCTR [79] | 13.16 | 20.63 | 20.55 | 18.11 | 38.94 | 15.40 | 25.27 | 22.01 | 20.89 | 36.03 |
| | DCPNet [80] | 16.52 | 20.20 | **33.61** | 23.44 | 36.57 | 16.97 | 25.08 | 25.10 | 22.38 | 35.71 |
| | NTRENet [81] | 15.51 | 19.65 | 30.07 | 21.74 | 33.68 | 15.18 | 23.50 | 31.83 | 23.50 | 32.16 |
| | DMNet [21] | 21.18 | 24.57 | 28.74 | 24.83 | **47.47** | 24.65 | 33.97 | **36.89** | 31.84 | **53.38** |
| | Ours | **22.97** | **27.79** | 28.14 | **26.30** | 46.83 | **26.29** | **34.88** | 36.84 | **32.67** | 52.88 |

In our model, the number of similarity maps in the MGS module (Section III-C) is set to 7, 13, and 30 for VGG-16, ResNet-50, and ResNet-101, respectively. For iSAID-$5^i$, the input image resolution is 256 × 256, with a batch size of 8 and a learning rate of 0.004. For LoveDA-$2^i$, the input image resolution is 473 × 473, with a batch size of 4 and a learning rate of 0.0005. We use the SGD optimizer with a momentum of 0.9 and a weight decay of 0.0001. The poly learning rate strategy from DeepLab [39] is adopted, and the model is trained for 200 epochs. The convolutional blocks in (8) and (10) consist of three convolutional layers with a kernel size of 3 and one convolutional layer with a kernel size of 1, with their weights being shared. The model employs data augmentation techniques, such as scaling, rotation, and Gaussian blur, to increase data diversity and enhance the model's generalization capabilities. The number of iterations in slot attention is set to 3. After training, we randomly selected 1000 pairs of support and query images from the test set to evaluate the model's performance.

In the $k$-shot setting, we need to retrain the model. Specifically, we average all similarity maps and prototypes to obtain a single set of similarity maps and prototypes. This way, the other modules of the model can operate in the same manner as in the one-shot setting.

### C. Comparison With SOTA

We compare our model with several leading FSS methods for natural scene analysis, including CyCTR [79], DCPNet [80], and NTRENet [81], as well as some methods designed for remote sensing scenes, such as SADMNet [32] and DMNet [21].

*1) iSAID-$5^i$:* The performance of our proposed method on the iSAID-$5^i$ dataset in both one-shot and five-shot settings is presented in Table I. From Table I, we can observe that our model achieves the highest mean mIoU performance across all backbone networks, establishing a new state of the art. Specifically, in the one-shot setting, the mean mIoU shows an improvement of 1.07% with VGG-16, 2.27% with ResNet-50, and 1.66% with ResNet-101 compared with DMNet [21]. Traditional methods, such as PFENet [16], CyCTR [79], and NTRENet [81], struggle to perform well on this dataset, as they do not consider the characteristics of remote sensing images. Our method considers the intrinsic characteristics of remote sensing images from a unified viewpoint and designs specific modules to address them, which helps improve performance. In the five-shot setting, our method continues to outperform the previous SOTA method DMNet [21] by a clear margin. Specifically, the mIoU increases by 1.35% with VGG-16, 2.53% with ResNet-50, and 1.69% with ResNet-101.

However, our model's advantage is not significant in terms of FB-IoU. Specifically, in the one-shot setting with VGG-16, our method underperforms DMNet [21] by 0.63%, despite achieving a higher mean mIoU. Meanwhile, in the one-shot setting with ResNet-50, we slightly outperform DMNet [21] by 0.03%, even though our method exceeds it by 2.27% in terms of mean mIoU. The reason lies in that foreground objects occupy only a small portion of the image compared with the complex background. Therefore, when the model is biased toward the background class, it often achieves higher results. This diminishes the persuasiveness of this metric.

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*2) LoveDA-$2^i$:* The performance of our proposed method on the LoveDA-$2^i$ dataset in both one-shot and five-shot settings is presented in Table II. As shown in Table II, the results of various models on this dataset are relatively lower. Both data distribution shifts and object size variations contribute to the difficulty of segmentation. However, our method still achieves the highest mean mIoU performance under both one-shot and five-shot settings across all backbone networks. This demonstrates our model's capability to handle complex segmentation scenarios. In particular, using ResNet-50 as the backbone network, the mean mIoU shows an improvement of 1.67% under the one-shot setting and 2.56% under the five-shot setting compared with DMNet [21]. It can also be observed that by increasing the support images from one-shot to five-shot, our method achieves a notable mean mIoU improvement of 5.56% with VGG-16, 5.95% with ResNet-50, and 6.33% with ResNet-101. The underlying reason may be that the LoveDA-$2^i$ dataset aligns more closely with the characteristics of remote sensing images that we have highlighted. As a result, on this dataset, our method can benefit from more support images to provide detailed support foreground subregions information and diverse backgrounds contexts, achieving a better match while filtering out irrelevant query features.

Similarly, on this dataset, our model's FB-IoU metric is less competitive. Our FB-IoU metric shows a decline compared with DMNet [21] under VGG-16 and ResNet-101, for reasons discussed in Section IV-C1.

*3) Model Complexity:* Model complexity is also an important criterion for evaluating a model. Therefore, we conduct a comparison experiment on model complexity. As shown in Table III, we compare the mean accuracy (mAcc), the size of learnable parameters, and the inference speed on a GeForce RTX 3090 GPU card under the ResNet-50 in the one-shot setting on the iSAID-$5^i$ dataset. Typically, the size of learnable parameters (MB) is obtained by summing all the learnable weights and biases, and inference speed (FPS) is achieved by calculating the number of images segmented per second by the model. In addition, mAcc is obtained by calculating the number of correctly classified pixels across threefold and then taking the average of them. From the table, we can observe that our model has 10.1 M learnable parameters (third fewest), with only a small gap of 2.7 M compared to the method with the fewest learnable parameters (DMNet [21]). Meanwhile, our method achieves an inference speed of 11.22 FPS (sixth fastest), with a noticeable gap (4.86 FPS) compared with the fastest method, PFENet [16]. The primary reason is that the MGS module needs to extract multi-granularity support and query feature maps to calculate the MGS matrices. However, our method achieves the best mAcc of 78.29, which indicates a satisfactory tradeoff among accuracy, parameters, and speed.

## D. Ablation Studies

All ablation experiments are conducted on the iSAID-$5^i$ dataset using the ResNet-50 network in the one-shot setting. In addition to the ablations on modules and hyperparameters, we also perform experiments under non-BAM setting to further validate the superiority of MGANet.

TABLE III
COMPARATIVE STUDY OF MODEL COMPLEXITY

| Method | mAcc | Params (MB) | Inference Speed (FPS) |
|---|---|---|---|
| SCL [18] | 75.71 | 11.9 | 10.59 |
| PFENet [16] | 75.88 | 10.8 | 16.08 |
| ASGNet [17] | 74.00 | 10.4 | 14.06 |
| CyCTR [79] | 76.11 | 7.5 | 10.36 |
| DCPNet [80] | 74.63 | 35.0 | 15.48 |
| NTRENet [81] | 76.38 | 20.8 | 13.83 |
| DMNet [21] | 77.47 | 7.4 | 12.58 |
| Ours | 78.29 | 10.1 | 11.22 |

TABLE IV
EFFECTS OF THE MGS AND AMPA MODULES. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREEFOLD COMPARED WITH THE BASELINE

| MGS | AMPA | fold0 | fold1 | fold2 | Mean | FB-IoU | △ |
|---|---|---|---|---|---|---|---|
| | | 54.45 | 40.68 | 53.60 | 49.58 | 64.46 | - |
| ✓ | | 54.98 | 42.82 | 55.08 | 50.96 | 63.84 | +1.38 |
| | ✓ | 55.16 | 42.29 | 54.90 | 50.78 | 63.96 | +1.20 |
| ✓ | ✓ | 55.81 | 43.52 | 56.23 | 51.85 | 64.49 | +2.27 |

TABLE V
RESULTS UNDER DIFFERENT ITERATIONS IN SLOT ATTENTION. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREEFOLD COMPARED WITH THE BASELINE

| Iteration | fold0 | fold1 | fold2 | Mean | FB-IoU | △ |
|---|---|---|---|---|---|---|
| 1 | 55.04 | 42.78 | 55.86 | 51.23 | 64.53 | +1.65 |
| 3 | 55.81 | 43.52 | 56.23 | 51.85 | 64.49 | +2.27 |
| 5 | 55.62 | 43.94 | 56.17 | 51.91 | 64.87 | +2.33 |

*1) Effects of Modules:* Table IV illustrates the results of ablation experiments demonstrating the effectiveness of each module we proposed. It can be observed that each module contributes positively to the overall performance on each fold. Specifically, the model with only MGS improves the mean mIoU performance by 1.38% compared with the baseline (the first row of the table). This indicates that incorporating support background information to calculate holistic correlation at multi-granularity helps to improve segmentation accuracy, as also discussed in Section IV-D7. Meanwhile, the model with only AMPA can outperform the baseline by 1.20%. This validates the feasibility of our approach to decompose the support foreground into more detailed subregions to achieve a desired match. Combining these two modules, our network achieves further performance improvement compared with the baseline, with an increase of 2.27% in mean IoU and 0.03% in FB-IoU. This demonstrates that our modules synergistically enhance each other, unleashing their full potential.

*2) Effects of Number of Iterations T:* Table V illustrates the impact of the number of iterations in the slot attention module on model accuracy. The model's accuracy improves significantly when we increase the number of iterations from 1 to 3. This improvement is attributed to the enhanced feature extraction capabilities of the slot attention module with more iterations. However, increasing the number of

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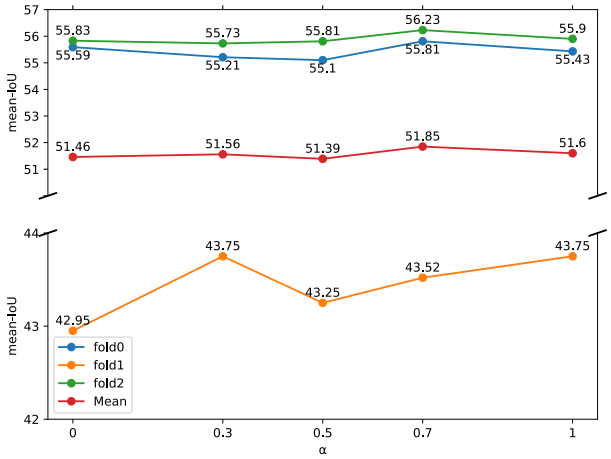



Fig. 6. mIoUs under different loss coefficients $\alpha$.

TABLE VI
PERFORMANCE COMPARISONS UNDER NON-BAM SETTINGS

| Method | fold0 | fold1 | fold2 | Mean | FB-IoU |
|---|---|---|---|---|---|
| $R^2$Net [84] | 41.22 | 21.64 | 35.28 | 32.71 | 63.86 |
| PCNet [85] | 40.24 | 24.64 | 31.31 | 32.06 | - |
| PCFNet [30] | 42.94 | 27.17 | 40.06 | 36.72 | 57.78 |
| HMRE [34] | 40.88 | 32.88 | 37.22 | 36.99 | - |
| $MS^2A^2$Net [28] | 34.96 | 25.30 | 30.71 | 30.32 | 59.81 |
| SCCNet [86] | 36.21 | 27.42 | 43.37 | 35.67 | - |
| Baseline | 37.75 | 26.03 | 42.14 | 35.31 | 56.11 |
| Ours | 39.90 | 30.72 | 44.18 | 38.27 | 54.32 |

iterations to 5 results in performance comparable to that with 3 iterations, indicating that the module's performance nearly saturates by the third iteration. To balance inference speed and accuracy, we ultimately set the number of iterations to 3.

*3) Effects of Loss Function Weights $\alpha$:* As shown in Fig. 6, we test the model's accuracy with different values of $\alpha$ in 0, 0.3, 0.5, 0.7, and 1.0 under the one-shot setting across various folds. In the figure, we do not display the other loss weight $\beta$, because we simply set $\beta$ to be 1-$\alpha$. The model achieves the best performance in terms of mIoU and FB-IoU when $\alpha$ is 0.7 and $\beta$ is 0.3.

*4) Effects of Non-BAM Setting:* To demonstrate the effectiveness of our proposed method, we also compare the performance of our model with other models under the non-BAM setting. In this setting, novel classes are labeled as background in the training images, which presents an additional challenge for the model to distinguish novel classes even when they are implicitly present in the training data. As shown in Table VI, our model achieves the best performance among these leading remote sensing FSS networks. Specifically, we achieve a mean IoU accuracy of 38.27%, surpassing the second-best method, HMRENet [34], by 1.28%, demonstrating the robustness and generalization ability of our approach under more challenging conditions.

*5) Effects of Out Channel Numbers:* We explore the impact of output channel numbers in the MGS module (see Section III-C) on model accuracy. Increasing the number of channels gives the MGS module greater influence in the decoder. While we have previously shown in Section IV-D1 that MGS improves model performance, simply increasing the proportion of MGS information does not necessarily lead to further improvements. This is because over-reliance on MGS can overshadow other critical information. As shown in Table VII, we evaluate the model's performance with different values of channel numbers in 32, 64, and 128. The model achieves the best performance when the output channel number is 64. Therefore, we set it to 64 for all experiments.

TABLE VII
RESULTS UNDER DIFFERENT OUTPUT CHANNEL NUMBERS. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREEFOLD COMPARED WITH THE BASELINE

| Number | fold0 | fold1 | fold2 | Mean | FB-IoU | △ |
|---|---|---|---|---|---|---|
| 32 | 54.98 | 43.67 | 55.66 | 51.43 | 64.78 | +1.85 |
| 64 | 55.81 | 43.52 | 56.23 | 51.85 | 64.49 | +2.27 |
| 128 | 55.24 | 42.99 | 55.93 | 51.39 | 64.24 | +1.81 |

TABLE VIII
RESULTS UNDER DIFFERENT ATTENTION MECHANISMS. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREEFOLD COMPARED WITH THE BASELINE

| Method | fold0 | fold1 | fold2 | Mean | FB-IoU | △ |
|---|---|---|---|---|---|---|
| cross attention | 54.37 | 42.88 | 56.46 | 51.24 | 64.52 | +1.66 |
| slot attention | 55.81 | 43.52 | 56.23 | 51.85 | 64.49 | +2.27 |

TABLE IX
RESULTS UNDER DIFFERENT BACKGROUND REMOVING METHODS. THE △ INDICATES THE CHANGE IN MEAN MIOU ACROSS THREEFOLD COMPARED WITH THE BASELINE

| Method | fold0 | fold1 | fold2 | Mean | FB-IoU | △ |
|---|---|---|---|---|---|---|
| pre-removing | 54.69 | 38.89 | 53.25 | 48.94 | 64.03 | -0.64 |
| post-removing | 55.81 | 43.52 | 56.23 | 51.85 | 64.49 | +2.27 |

*6) Effects of Slot Attention:* To evaluate the contribution of slot attention [35] to our model, we conduct experiments comparing it with cross attention. The results are shown in Table VIII. We can observe slot attention outperforms cross attention, achieving an improvement in mean mIoU accuracy by 0.61%.

*7) Effects of Postremoving:* To validate the superiority of our postremoving strategy, we compare it with the preremoving strategy. As shown in Table IX, our strategy achieves a greater performance improvement, surpassing the preremoving strategy by 2.91% (+2.27% versus −0.64%). Moreover, the preremoving strategy even performs worse performance than the baseline, indicating that it produces lower quality similarity maps and causes the decoder to focus on nontarget regions. In addition, we present visual results in Fig. 8. These heatmaps generated by our postremoving strategy more accurately highlight the query target locations.

*8) Effects of Different Backbones:* We further compare the impact of different backbone networks on the experimental results. In this section, we use the base version of the Swin Transformer, which consists of four Swin Transformer layers with the depths of (2, 2, 18, 2). We utilize the feature maps from the last 13 layers of the third stage (18 layers) to

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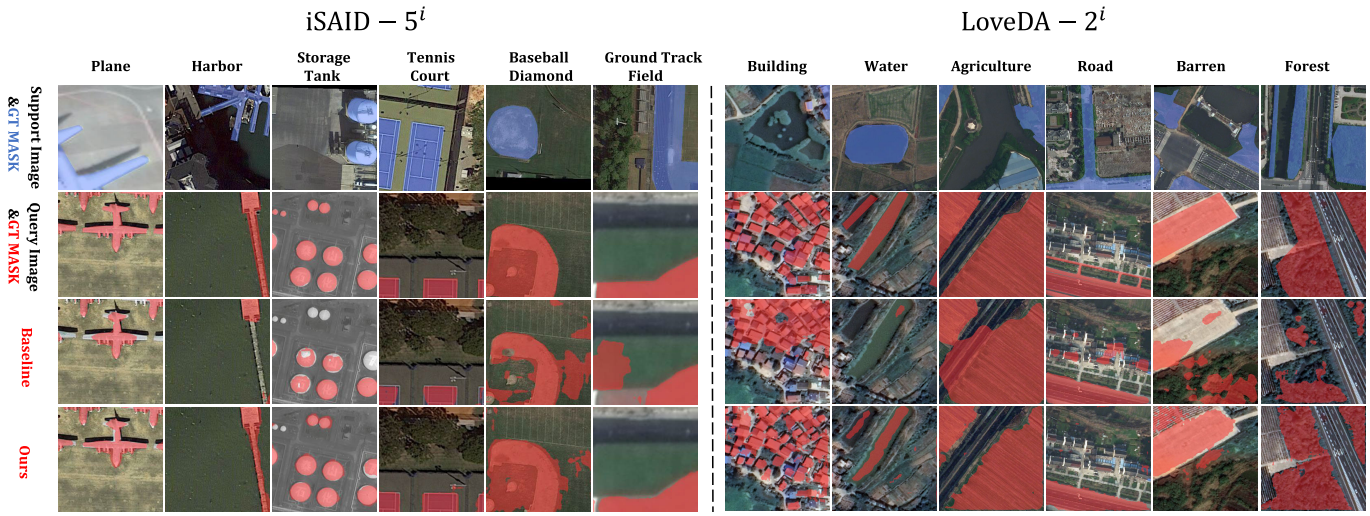



Fig. 7. Comparison of model segmentation performance on iSAID-$5^i$ and LoveDA-$2^i$ in one-shot Setting. Each row, from top to bottom, shows the support image and its ground truth, the query image and its ground truth, the segmentation result from the baseline model, and the segmentation result from our model.

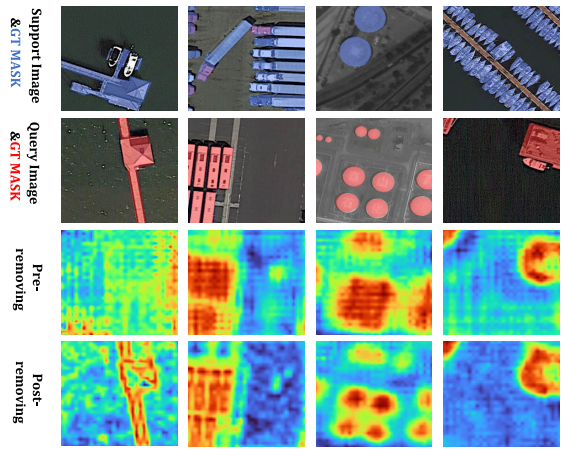



Fig. 8. Comparison of heatmaps generated by MGS module and preremoving method. Each row, from top to bottom, shows the support image and its ground truth, the query image and its ground truth, the heatmap generated by preremoving method, and the heatmap generated by the MGS module.

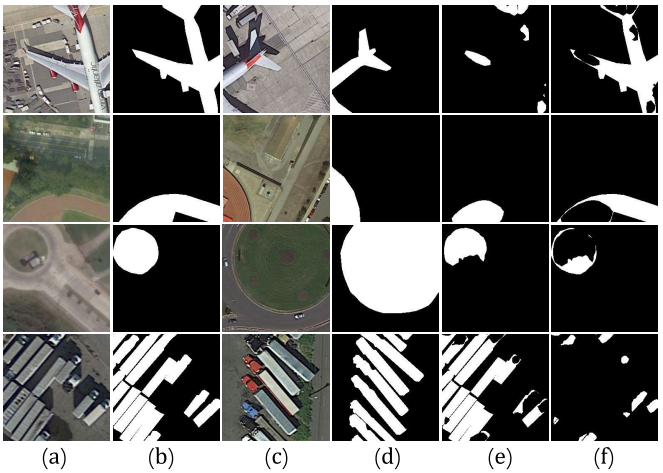



Fig. 9. Qualitative analysis results. (a) Support image. (b) Support ground-truth mask. (c) Query image. (d) Query ground-truth mask. (e) $M^{\text{main}}$ in (11). (f) $M^{\text{aux}}$ in (11).

TABLE X
RESULTS UNDER DIFFERENT BACKBONE NETWORKS

| Backbone | fold0 | fold1 | fold2 | Mean | FB-IoU |
|---|---|---|---|---|---|
| VGG-16 | 49.55 | 40.47 | 48.72 | 46.25 | 62.47 |
| ResNet-50 | 55.81 | 43.52 | 56.23 | 51.85 | 64.49 |
| ResNet-101 | 55.03 | 42.90 | 54.69 | 50.87 | 64.35 |
| Swin | 49.88 | 39.50 | 49.44 | 46.27 | 59.72 |

compute the multiscale similarity map, while the final two layers are used to compute the intermediate features for the support and query images. During training, we set the learning rate to 0.0002, while keeping all other settings unchanged. From Table X, we find that using the Swin Transformer as the backbone achieves only a slight improvement in mIoU accuracy compared with VGG-16. This indicates that the convolutional-style networks seem to be more suitable for the remote sensing FSS task under our MGANet framework.

*9) Visualizations:* Fig. 7 presents the examples of prediction results in the one-shot setting on the iSAID-$5^i$ and LoveDA-$2^i$ datasets. It can be observed from the figure that our model outperforms the baseline across all scenarios. In the "building" category, due to the small and scattered distribution of objects, the baseline's predictions for different instances tend to merge without clear boundaries, whereas our model can distinguish the boundaries effectively. In cases where the baseline predictions are notably poor (e.g., "harbor" and "water" categories), our model still achieves results that are close to the ground truth.

To better understand the AMPA module, we provide several visualizations in Fig. 9. In the first row of the figure, we can observe that in the query image, only the tail and wings of the plane are visible, while the support image shows almost the entire plane. This indicates a significant intraclass inconsistency between the support and query images. In this case, AMPA successfully decomposes the wings and tail

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from the plane in the support image (see the fifth column) and matches the regions in $M^{\text{main}}$, which has high intraclass consistency with the query, leading to improved accuracy. Similar observations can be made in the second to fourth rows.

## V. Conclusion

In this article, we propose a multi-granularity aggregation network to address FSS in remote sensing, which mainly consists of MGS and AMPA modules. Specifically, the MGS module locates the target regions in the query by computing the holistic correlation using multi-granularity paired support and query features. By incorporating support background information to compute holistic correlations among support and query features, MGS achieves superior performance compared with methods that only focus on the relationship between the support foreground and the query foreground. In addition, to alleviate the intraclass inconsistency of support and query images, the AMPA module decomposes the support foreground region into a mainstay and auxiliary subregions and utilizes slot attention to enhance query features. Our method achieves state-of-the-art performances on the iSAID-$5^i$ and LoveDA-$2^i$ datasets.

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